

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

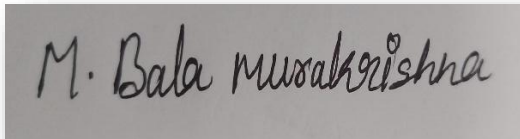
COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Code of Conduct:

I MADDIRALA BALAMURALI KRISHNA (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Duration: 1 Hour

A rectangular box containing a handwritten signature in black ink. The signature reads "M. Bala Murali Krishna" in a cursive script.

MADDIRALA BALAMURALI KRISHNA

Signature of student

Name of student

QUESTIONS: 1

Total Marks:50

Annexure A

Error Analysis Task

Question 1 – Error Analysis Task

A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

a) Identify the error – what the gap between training and testing MSE indicates

- The reported metric is Mean Squared Error (MSE), so lower values are better; here the model's training error (0.85) is roughly 6x smaller than its testing error (5.20), which is a large and asymmetric gap rather than normal noise between two data splits.
- This pattern — low error on data the model has seen, sharply higher error on data it has not seen — is the textbook signature of overfitting: the network has fit the training set (including its noise and idiosyncrasies) rather than learning the underlying score-prediction relationship that generalizes to new students.
- It is not simply that the model is bad, because a genuinely under-capacity or poorly-trained model would show high error on both splits; the fact that training error is very low rules out underfitting and points specifically to a generalization gap between train and test performance.
- A secondary possibility worth ruling out is a train/test distribution mismatch or preprocessing error (e.g., features scaled using inconsistent statistics, or the test set drawn from a different population of students) — this would also inflate test MSE even if the model itself were not overfitting in the classical sense.

b) Analyze why this error occurs (reasoning)

- Excess model capacity relative to data size: if the network has many parameters (deep/wide layers) relative to the number of training examples, it has enough degrees of freedom to memorize individual training samples — including their noise — instead of learning the general trend that predicts unseen students' scores.
- Insufficient or non-representative training data: with too few samples, or samples that do not span the same range/distribution as the test set, the model's learned mapping is only reliable near the training points and extrapolates poorly elsewhere, showing up as a large jump in test MSE.
- Lack of regularization: without a mechanism to constrain weight magnitudes or effective capacity (L1/L2 penalty, dropout, early stopping), gradient descent is free to drive training loss arbitrarily low by fitting noise, which necessarily hurts generalization.
- Training for too many epochs: the classic overfitting curve shows training loss continuing to fall while validation/test loss falls initially and then rises again — continuing to train well past this turning point widens exactly the kind of gap seen here.
- Feature/target scaling or preprocessing inconsistency: if normalization statistics (mean/std, min/max) were computed on the training set and then wrongly recomputed — or not applied at all — on the test set, the test inputs the network sees are effectively out-of-distribution, inflating test error independent of overfitting.
- Small or unrepresentative test set: if the test set is small, a handful of hard-to-predict students (outlier scores) can disproportionately raise the MSE, since squared error penalizes large individual errors heavily — this can exaggerate the apparent gap beyond the true generalization error.

c) Suggest corrective measures

- Add regularization: apply L2 (ridge) weight decay and/or Dropout on hidden layers to constrain the effective capacity of the network and discourage it from fitting training-set noise.
- Use early stopping: monitor validation MSE during training and stop (or restore the best checkpoint) once validation error starts rising, even though training error would keep falling — this directly targets the train/test divergence point.
- Simplify the architecture: reduce the number of hidden layers/neurons if the network is heavily over-parameterized for the size of the dataset, lowering its capacity to memorize.
- Increase and diversify training data: collect more labelled student-score examples, or use data augmentation where appropriate, so the training distribution better represents the population the model will be tested on.

- Apply k-fold cross-validation: rather than a single train/test split, use cross-validation to get a more reliable estimate of generalization performance and to select hyperparameters (learning rate, regularization strength, architecture size) based on held-out performance, not training performance.
- Verify and fix preprocessing: ensure feature scaling/normalization statistics are computed only on the training set and then applied unchanged to the test set, and confirm there is no data leakage between splits.
- Use Batch Normalization: stabilizes and regularizes the learned representations across layers, which often narrows the train–test performance gap as a side effect of more stable training.

d) Interpret the results – impact on model performance

- The 0.85 vs 5.20 MSE gap means the network is not yet reliable for deployment: its apparently excellent training performance is misleading, since real-world (test-time) predictions of student scores would be, on average, considerably less accurate than the training numbers suggest.
- Because MSE penalizes large errors quadratically, a test MSE of 5.20 implies the model is making some substantially larger individual mistakes on unseen students than the training numbers indicate — possibly over- or under-predicting certain groups of students, which is a reliability concern if scores are used for decisions.
- Practically, any conclusions drawn from the model's training-time metric alone (0.85 MSE) would be overly optimistic; the test MSE (5.20) is the more honest estimate of real-world performance and should be the number reported and monitored going forward.
- After applying the corrective measures above, the expected outcome is that the training MSE will rise slightly (since the model is no longer allowed to memorize freely) while the test MSE falls, so the two numbers converge toward each other — a small, healthy gap is the sign that the model has learned generalizable patterns rather than memorized the training set.